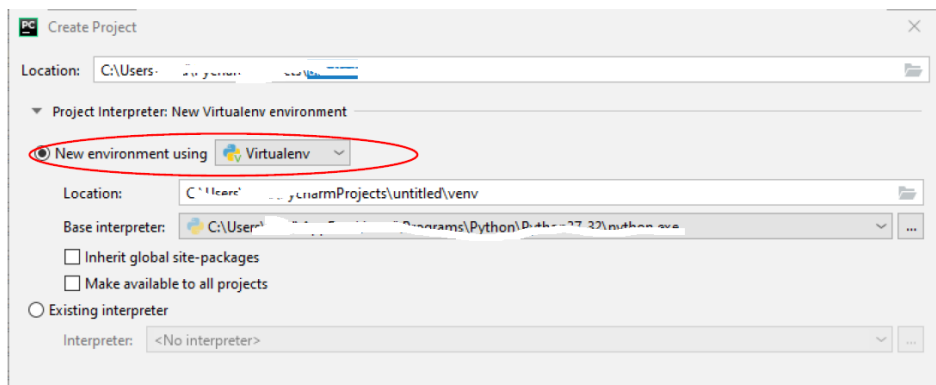


Installation :

- Create an account in Github and Heroku
- Git : <https://git-scm.com/download/win>
- Heroku CLI : <https://devcenter.heroku.com/articles/heroku-cli#download-and-install>
- In the Lab folder, you will find the pycharm IDE .exe. Launch it and make it choose a virtual environment (see figure below)



- Open pycharm and install flask using either the “pip install flask” command in the terminal or by searching for flask in the “Python Packages” tab

Example 1 :

To test the flask installation, create a new project in which you create a "hello.py" file and then run it.

```
from flask import Flask
app = Flask(__name__)

@app.route('/')
def hello_world():
    return 'Hello World'

if __name__ == '__main__':
    app.run()
```

You need to see “Hello World” by visiting this link : <http://127.0.0.1:5000/>

Example 2:

Create another project and then create:

- A file named “app.py” containing the code below:

```
from flask import Flask, render_template
app = Flask(__name__)

@app.route('/result')
```

```
def result():
    dict = {'phy':50,'che':60,'maths':70}
    return render_template('result.html', result = dict)

if __name__ == '__main__':
    app.run(debug = True)
```

- Another html file named “result.html” in a folder (to be created) named “templates” containing the code below:

```
<!doctype html>
<html>
  <body>
    <table border = 1>
      {% for key, value in result.items() %}
        <tr>
          <th> {{ key }} </th>
          <td> {{ value }} </td>
        </tr>
      {% endfor %}
    </table>
  </body>
</html>
```

Then run this application and see the result you get.

Example 3:

Create another project and then create:

- A file named “app.py” containing the code below:

```
from flask import Flask, render_template, request
app = Flask(__name__)

@app.route('/')
def student():
    return render_template('student.html')

@app.route('/result', methods = ['POST', 'GET'])
def result():
    if request.method == 'POST':
        result = request.form
        return render_template("result.html", result = result)

if __name__ == '__main__':
    app.run(debug = True)
```

- In the templates folder (to be created), an html file named “student.html”, containing the code below:

```

<html>
  <body>
    <form action = "http://localhost:5000/result" method = "POST">
      <p>Name <input type = "text" name = "Name" /></p>
      <p>Physics <input type = "text" name = "Physics" /></p>
      <p>Chemistry <input type = "text" name = "chemistry" /></p>
      <p>Maths <input type = "text" name = "Mathematics" /></p>
      <p><input type = "submit" value = "submit" /></p>
    </form>
  </body>
</html>

```

- In the same templates folder, an html file named “result.html”, containing the code below:

```

<!doctype html>
<html>
  <body>
    <table border = 1>
      {% for key, value in result.items() %}
        <tr>
          <th> {{ key }} </th>
          <td> {{ value }} </td>
        </tr>
      {% endfor %}
    </table>
  </body>
</html>

```

Then run this application and see the result you get.

Deploying a machine learning model on the web

Open the “furniture_prediction” project, and then open the “app.py” and “index.html” files and look at their contents.

Then open the “Furniture prediction notebook” and run it then put the generated .pkl file in the project folder.

Run the project and do some manipulations.

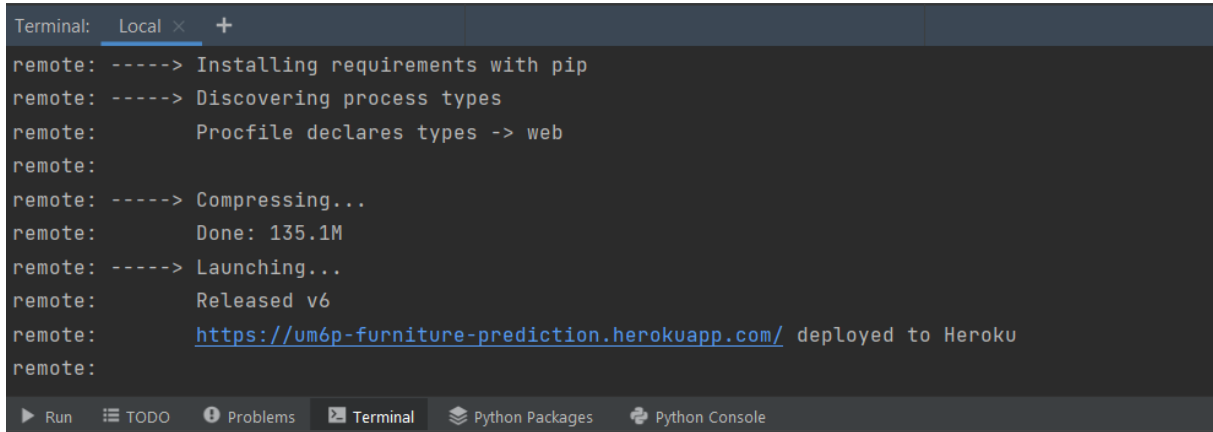
Deployment on the Heroku platform

Open the terminal in Pycharm and type the following commands:

1. heroku login
2. Creating a Heroku app: heroku create furniture-prediction-app --region eu.
 - a. To view the applications created: heroku apps
3. git init

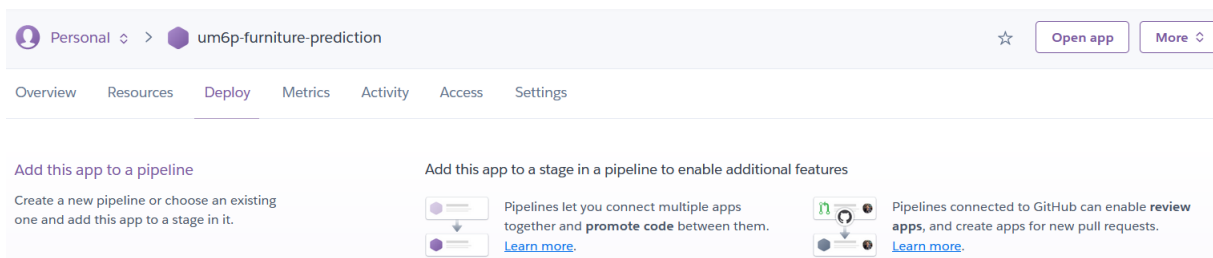
4. `git add .`
5. `git commit -am "initial change"`
6. `git push heroku master`

Click on link which will appear in the terminal



```
Terminal: Local x +
remote: -----> Installing requirements with pip
remote: -----> Discovering process types
remote:      Procfile declares types -> web
remote:
remote: -----> Compressing...
remote:      Done: 135.1M
remote: -----> Launching...
remote:      Released v6
remote:      https://um6p-furniture-prediction.herokuapp.com/ deployed to Heroku
remote:
```

Or by opening the application on Heroku.



Assignment:

1. Choose a dataset, containing tabular data, of your choice
 - <https://www.kaggle.com/datasets>
 - <https://datasetsearch.research.google.com/>
2. Prepare your data (treatment of missing values, etc.)
3. Make sure to bring out the most relevant characteristics
4. Use these characteristics to create the prediction page (like in the example above)
5. Compare the performance of multiple ML models to keep the best one at the end.
6. Create an application (i.e., a form) that predicts the target (value or class) of a record.